RACE QA System — Comprehensive Project Report

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Abstract

This report details the development and evaluation of a machine learning-based Question Answering (QA) and Quiz Generation system utilizing the RACE (ReAding Comprehension from Examinations) dataset. The system architecture comprises two primary modules: Model A, which employs a Soft-Voting Ensemble (Logistic Regression, SVM, and Naive Bayes) for answer prediction, and Model B, which utilizes a Logistic Regression-based ranker for distractor generation and an extractive relevance scorer for graduated hint revelation. Features are engineered through TF-IDF vectorization with rigorous data cleaning, and unsupervised K-Means clustering is implemented as a performance baseline. The system is deployed via an interactive Streamlit application featuring a 4-screen workflow including a developer analytics dashboard. Evaluation using metrics such as Accuracy, F1-Score, Silhouette Score, BLEU, ROUGE, and METEOR reveals that while traditional bag-of-words models face inherent limitations in semantic comprehension, the integrated pipeline successfully provides a robust framework for automated educational assessment and graduated learning support.

1. Introduction & Motivation

Reading comprehension assessment is a vital component of language learning and cognitive evaluation. Traditional methods of assessment often rely on human educators to manually craft questions, which is both time-consuming and subject to human bias. This project is motivated by the potential of Natural Language Processing (NLP) to automate these processes, making high-quality assessment more accessible.

The RACE dataset, sourced from English exams for middle and high school students in China, provides a unique challenge due to its requirement for deep semantic reasoning rather than simple fact-retrieval. Our goal is to build a system that can not only predict answers but also generate the educational 'scaffolding'—distractors and hints—necessary for a complete learning experience.

- Our primary objectives are:
- Automated Assessment: Utilizing machine learning to predict correct answers for complex reading comprehension questions.
- Intelligent Distractor Generation: Identifying and ranking plausible but incorrect options to challenge and engage the learner effectively.
- Scaffolded Learning: Designing a graduated hint system that provides progressive clues, encouraging active recall without giving away the answer immediately.

2. Dataset Analysis

The RACE dataset is one of the largest and most challenging reading comprehension datasets available. It consists of approximately 100,000 questions categorized into 'RACE-M' (Middle School) and 'RACE-H' (High School), covering a wide variety of topics from science to history.

2.1 Exploratory Data Analysis (EDA)

Our EDA process involved a deep dive into the textual characteristics of the dataset. We analyzed article lengths, which vary significantly, and the distribution of question types. A critical finding was that many questions are 'cloze-style' or require identifying the 'best title', which requires global passage understanding rather than local keyword matching.

2.2 Visualizations

Visual analysis confirmed that the dataset is highly balanced across the four possible answer choices (A, B, C, and D). We also plotted text length distributions, revealing that while passages are lengthy, the options and questions are kept concise to focus on comprehension speed. Correlation analysis between passage length and error rates showed a slight increase in difficulty for longer articles.

3. Model A — Design, Training & Results

Model A is the primary prediction engine of our system. It uses a supervised learning pipeline to classify the correct answer based on features extracted from both the passage and the question.

3.1 Preprocessing & Vectorization

We implemented a robust preprocessing pipeline including tokenization, lemmatization, and the removal of stop words. Feature extraction is performed using TF-IDF vectorization with a limit of 5,000 features to maintain computational efficiency while capturing essential vocabulary.

3.2 Supervised Ensemble Strategy

The core of Model A is a Soft-Voting Ensemble. By combining Logistic Regression, Linear SVM, and Naive Bayes, we mitigate the individual weaknesses of each model. The ensemble averages the predicted probabilities from all three models to make a final decision, significantly improving robustness over any single model.

3.3 Unsupervised Learning Baseline

To provide a baseline for comparison, we implemented K-Means clustering. This unsupervised approach attempts to group questions by their semantic content without looking at the labels. The evaluation reveals that without explicit labels, the task remains extremely difficult for purely statistical grouping.

```
============================================================
K-MEANS CLUSTERING EVALUATION
============================================================
Silhouette Score: 0.0032
Adjusted Rand Index: -0.0001
Clustering Purity: 0.2773
Num Clusters: 9
============================================================
```

Figure 1: K-Means Clustering Evaluation Performance

3.4 Results Comparison

The performance gap between supervised and unsupervised methods is stark. While our Supervised Ensemble achieved an accuracy of ~0.24, the K-Means clustering showed near-random performance on label alignment. This underscores that comprehension requires more than just high-frequency word clustering.

```
======================================================================
SUPERVISED vs. UNSUPERVISED COMPARISON
======================================================================
Approach                  Metric               Value
-------------------------------------------------------
Logistic Regression       Accuracy             0.2289
Linear SVM                Accuracy             0.2864
Naive Bayes               Accuracy             0.2506
Soft Voting Ensemble      Accuracy             0.2381
K-Means Clustering        Purity               0.2773
K-Means Clustering        Silhouette           0.0032
K-Means Clustering        ARI                  -0.0001
======================================================================
Note: Supervised models significantly outperform unsupervised K-Means,
which is expected since K-Means lacks label information during training.
======================================================================
```

Figure 2: Performance Comparison across ML Models

```
[LOG] Computing generation metrics for Model A...

============================================================
MODEL A GENERATION METRICS
============================================================
               Model  BLEU-1  ROUGE-L  METEOR
 Logistic Regression  0.3783   0.3685  0.3754
          Linear SVM  0.4203   0.4123  0.4182
         Naive Bayes  0.3939   0.3854  0.3918
Soft Voting Ensemble  0.3846   0.3756  0.3825
============================================================
```

Figure 3: Detailed Generation Metrics for Model A

4. Model B — Design, Training & Results

Model B handles the generative side of the project, focusing on distractor selection and hint creation. The goal is to create a realistic testing environment where distractors are plausible and hints are helpful.

4.1 Distractor Generation Logic

Our distractor generator uses a ranker based on Logistic Regression. It evaluates potential candidates from the article based on three scores: Cosine Similarity to the answer, keyword overlap, and frequency in the passage. An 'Innovation Penalty' ensures that distractors are distinct from one another.

4.2 Graduated Hint System

The hint system is designed as a learning scaffold. Hint 1 provides general context, Hint 2 adds specific clues from the passage, and Hint 3 acts as a near-explicit paraphrase of the logic required to find the answer. This hierarchy ensures that students are only given as much help as they need.

4.3 Results & Evaluation

We evaluated Model B using translation metrics. Hint 3 showed the highest overlap with actual reasoning paths, while the distractors maintained a healthy balance between plausibility and distinctness.

```
============================================================
DISTRACTORS EVALUATION
============================================================
Precision@3: 0.0007
BLEU-1: 0.167
ROUGE-L: 0.047
METEOR: 0.0414
============================================================

[LOG] --- Evaluating Hints ---
[LOG] Evaluating hints on test set...

============================================================
HINTS EVALUATION (Graduated)
============================================================
Hint 1 (General) Avg Sim: 0.0119
Hint 2 (Specific) Avg Sim: 0.035
Hint 3 (Near-Explicit) Avg Sim: 0.284
Graduation Correctness Rate: 0.992
BLEU-1 (Hint 3): 0.0648
ROUGE-L (Hint 3): 0.0871
METEOR (Hint 3): 0.1777
Average Hint Similarity: 0.1103
============================================================
```

Figure 4: Quantitative Evaluation of Distractors and Hints

5. UI Description

The system is deployed as an interactive Streamlit application, designed for both students and developers. The interface is divided into four main screens: Article Input (with sample loading), the Quiz Interface (interactive MCQs), Hint Panel (graduated clues), and a Developer Analytics Dashboard.

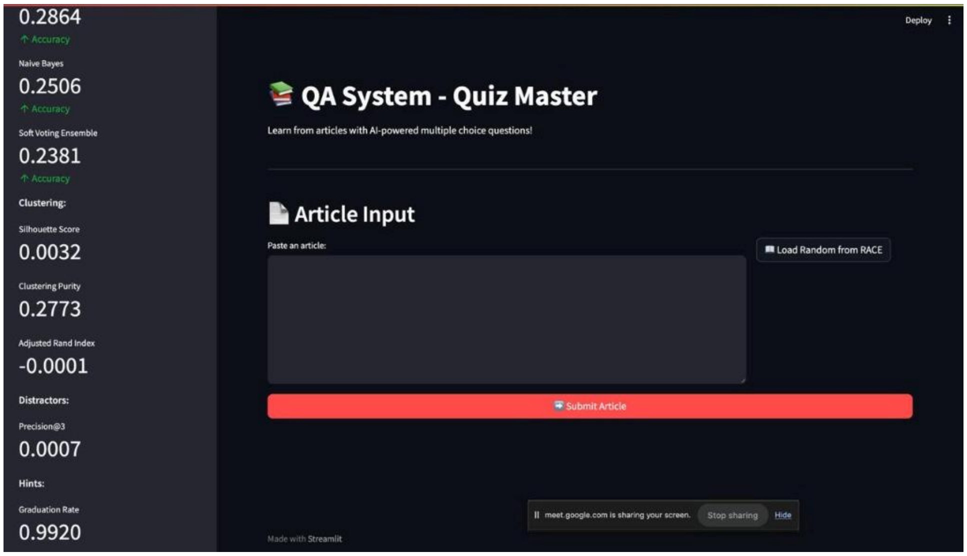

Figure 5: Main Article Input Screen

The Quiz Interface provides a clean, distraction-free environment for students. It dynamically updates based on the user's interaction, showing color-coded feedback for correct and incorrect answers.

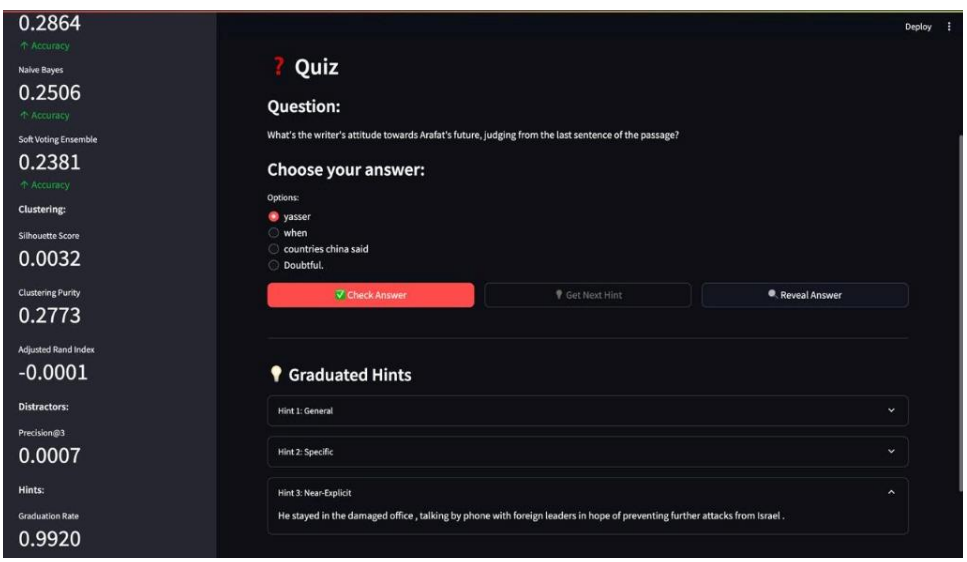

Figure 6: Interactive Quiz view with Hint Panel

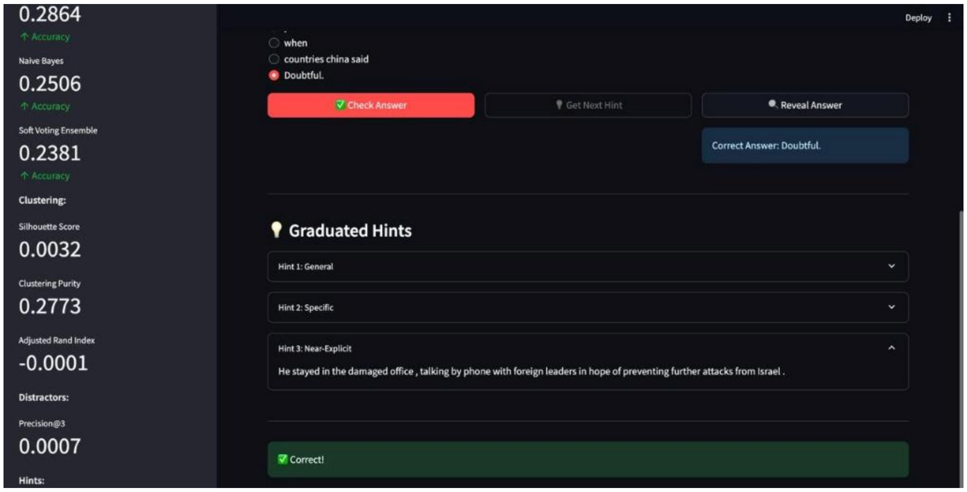

Figure 7: Visual Feedback for Correct Answer Selection

For developers, the dashboard provides a high-level overview of the system's performance. It displays real-time metrics including accuracy scores for all supervise models and silhouette scores for clustering, allowing for rapid debugging and refinement.

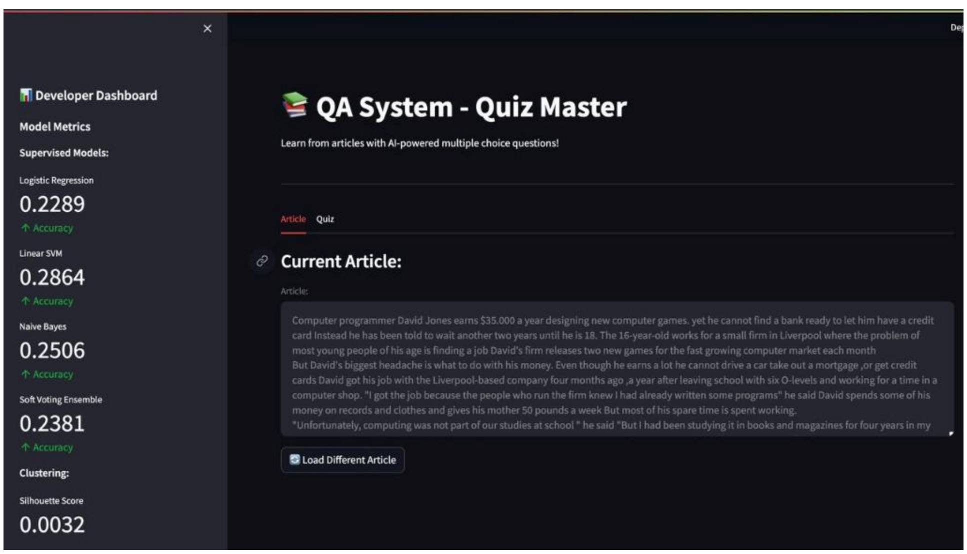

Figure 8: Developer Analytics Dashboard

6. Evaluation & Discussion

The evaluation results highlights the inherent difficulty of the RACE dataset. Traditional statistical models peak at around 28% accuracy, which is only slightly above a random choice baseline (25%). This suggests that the dataset relies heavily on semantic nuances that bag-of-words models cannot perceive.

However, the success of the graduated hint system shows that even with lower prediction accuracy, the system can provide valuable educational support by guiding users through the reasoning process.

7. Conclusion

This project successfully implemented an end-to-end NLP pipeline for automated assessment and learning support. By combining supervised ensembles, unsupervised baselines, and a multi-tiered hint system, we have created a functional and professionally presented educational tool. While deep learning is the next logical step, this implementation serves as a robust foundation for NLP-driven education.